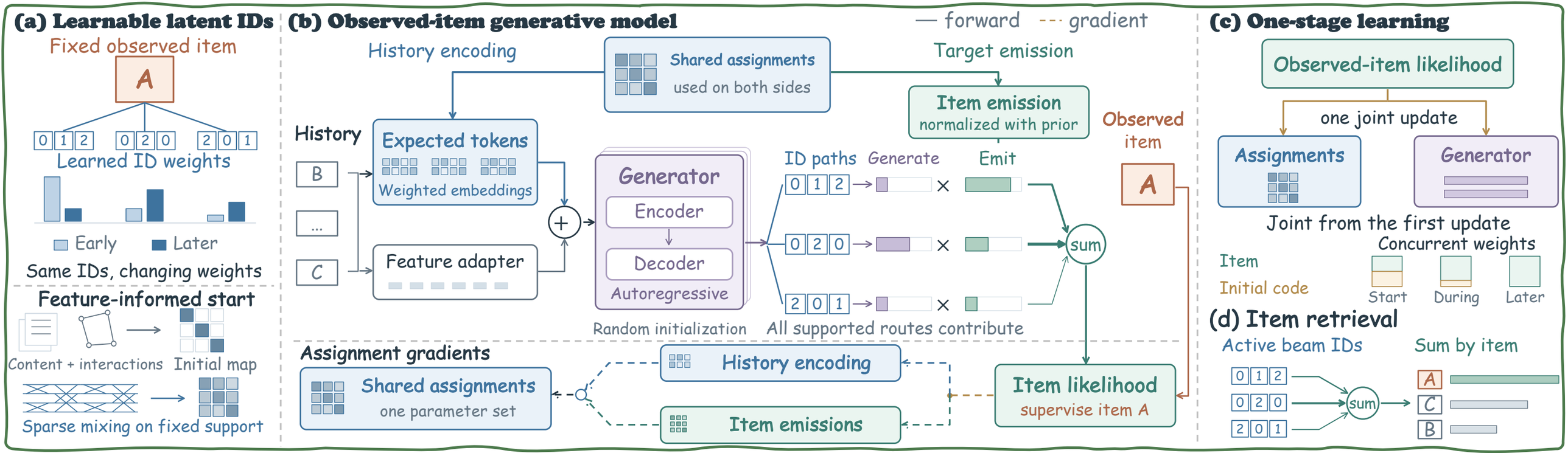

(a) Learnable latent IDs
Fixed observed item
A
0 1 2
0 2 0
2 0 1
Learned ID weights
Early
Later
Same IDs, changing weights
Feature-informed start
Content + interactions
Initial map
Sparse mixing on fixed support
(b) Observed-item generative model
forward
gradient
History encoding
Shared assignments
used on both sides
Target emission
Item emission
normalized with prior
Observed item
A
History
B
...
C
Expected tokens
Weighted embeddings
Feature adapter
Generator
Encoder
Decoder
Autoregressive
Random initialization
ID paths
Generate
Emit
0 1 2
0 2 0
2 0 1
sum
All supported routes contribute
Assignment gradients
Shared assignments
one parameter set
History encoding
Item emissions
Item likelihood
supervise item A
(c) One-stage learning
Observed-item likelihood
one joint update
Assignments
Generator
Joint from the first update
Concurrent weights
Item
Initial code
Start
During
Later
(d) Item retrieval
Active beam IDs
Sum by item
0 1 2
0 2 0
2 0 1
sum
A
C
B